

AI BASED INTEGRATED SOLUTION FOR SMART PEPPER FARMING IN SRI LANKA

25-26J-479

Project Proposal Report

Ranasinghe I.C

B.Sc. (Hons) Degree in Information Technology Specialized in Software
Engineering

Department of Computer Science and Software Engineering

Sri Lanka Institute of Information Technology
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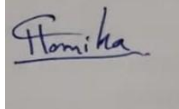
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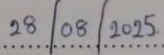
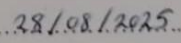
August 2025

DECLARATION

I declare that this is my own work, and this proposal does not incorporate without acknowledgment any material previously submitted for a degree or diploma in any other university or Institute of higher learning, and to the best of my knowledge and belief, it does not contain any material previously published or written by another person except where the acknowledgment is made in the text.

Name	Student ID	Signature
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The above candidate is carrying out research for the undergraduate dissertation under my supervision.

 Signature of the supervisor (Ms. Ishara Weerathunga)	 Date
 Signature of the co-supervisor (Mr. Eishan Weerasinghe)	 Date

ABSTRACT

Black pepper cultivation faces significant threats from various leaf diseases, which can severely affect yield and quality. This research aims to enhance disease detection and spread estimation in black pepper plants by leveraging artificial intelligence (AI), image processing, and deep learning techniques. The primary focus is on analyzing leaf images to identify infections and estimate the percentage of disease spread across the plant.

The proposed methodology involves three main tasks: developing a comprehensive dataset of healthy and diseased black pepper leaves under varying environmental conditions, applying image processing techniques to enhance and segment leaf features, and training a Convolutional Neural Network (CNN) model for accurate disease classification and spread estimation. The system will also integrate environmental parameters such as temperature, humidity, and soil moisture to improve prediction accuracy.

This research uniquely emphasizes not only classifying the type of disease but also quantifying the extent of infection, which is often overlooked in existing studies. The CNN model will be optimized to address challenges such as varying lighting conditions, leaf overlaps, and partial visibility. Data will be collected from both controlled environments and real field conditions to ensure robustness.

To validate the system's performance, predictions will be benchmarked against expert agricultural assessments. The outcome of this study is expected to assist farmers and agricultural experts by providing an automated, accurate, and real-time disease monitoring tool, ultimately supporting better disease management and improving crop productivity.

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1. INTRODUCTION

In recent years, precision agriculture has emerged as a critical tool in ensuring sustainable crop production and improving food security. Among various crops, black pepper (*Piper nigrum* L.) holds significant economic importance as a globally traded spice. However, black pepper cultivation faces severe threats from leaf diseases such as *Phytophthora foot rot*, anthracnose, and yellow mottle disease. These infections not only reduce yield but also affect the quality of the produce, resulting in considerable economic losses for farmers. Early detection and accurate assessment of disease spread are essential to prevent large-scale outbreaks and to implement timely control measures.

This study proposes a system that leverages artificial intelligence (AI), image processing, and deep learning techniques to detect black pepper leaf diseases and quantify the percentage of infection spread across the plant. By utilizing computer vision models—particularly CNNs—the system aims to identify visible symptoms such as lesions, discoloration, and fungal growth with high accuracy. Environmental parameters including temperature, humidity, and soil moisture will also be integrated into the analysis to improve disease prediction reliability.

The proposed system will employ a dedicated image dataset collected from both healthy and diseased black pepper leaves under varying conditions. Advanced preprocessing techniques will enhance image clarity, segment the affected areas, and remove noise. These processed images will then be analyzed to classify disease types and estimate infection severity. By mapping disease progression data with environmental trends, the system will provide valuable insights into potential outbreak patterns.

Beyond simple classification, this research focuses on quantifying the extent of infection on each leaf and extrapolating these results to estimate whole-plant infection spread. This capability, often overlooked in existing literature, will enable farmers and agricultural officers to assess the urgency of interventions and prioritize disease management strategies. Furthermore, the use of AI-driven models reduces reliance on subjective human judgment, offering more consistent and objective evaluations.

By developing an accurate and automated disease detection and analysis framework, this research has the potential to transform black pepper disease management. It will empower stakeholders with real-time, data-driven decision-making tools, ultimately contributing to higher crop yields, reduced pesticide usage, and more sustainable agricultural practices.

1.1 Background and Literature Survey

Black pepper (*Piper nigrum* L.) is one of the most valuable spice crops worldwide, cultivated extensively in tropical regions such as Sri Lanka, India, Vietnam, and Indonesia. Its high economic value makes it a vital source of income for smallholder farmers. However, the crop is highly susceptible to fungal, bacterial, and viral diseases such as *Phytophthora foot rot*, anthracnose, leaf blight, and yellow mottle virus. These diseases can significantly reduce both yield and quality, leading to economic losses for farming communities.

Early detection is crucial for effective management, as black pepper diseases often spread rapidly under favorable environmental conditions—high humidity, warm temperatures, and poor soil drainage. Traditional detection methods rely on visual inspection by agricultural officers or farmers, which, while practical, is time-consuming, labor-intensive, and often subjective. Early-stage symptoms can be subtle and difficult to differentiate from natural leaf aging or nutrient deficiencies, making timely intervention challenging.

Advancements in precision agriculture, image processing, and artificial intelligence (AI) have opened new possibilities for automated disease detection. Computer vision techniques, especially Convolutional Neural Networks (CNNs), have demonstrated high accuracy in identifying plant diseases by analyzing changes in leaf texture, color, and shape. Studies by Barbedo (2018) and Ferentinos (2018) showed that CNN models could classify plant diseases with over 90% accuracy from raw leaf images, eliminating the need for extensive manual feature engineering. However, research on black pepper diseases remains limited compared to other crops such as rice, tomato, and wheat, and most existing studies focus only on disease type classification rather than quantifying severity or spread.

One promising enhancement for plant health assessment is the use of **spectral vegetation indices**, particularly the **Normalized Difference Vegetation Index (NDVI)**. NDVI is a widely used remote sensing metric that evaluates plant health by measuring differences between near-infrared (NIR) and red light reflectance. Healthy vegetation reflects more NIR and absorbs more red light, resulting in higher NDVI values, whereas diseased or stressed plants show lower NDVI values. By integrating NDVI measurements with image-based disease detection, researchers can capture physiological changes in plants—such as reduced chlorophyll content—before visible symptoms appear. This makes NDVI

especially valuable for early disease detection and monitoring crop recovery after treatment.

Recent agricultural studies have successfully combined NDVI with machine learning models to improve disease detection accuracy and predict yield losses. For example, multispectral cameras mounted on drones or handheld NDVI sensors have been used to scan fields and generate vegetation health maps, allowing farmers to identify and prioritize treatment zones. In the context of black pepper, NDVI-based scanning can provide a complementary layer of data to CNN-based image analysis, enabling both early detection and severity quantification at the whole-plant or field level.

Additionally, modern image segmentation techniques such as U-Net and Mask R-CNN can be applied alongside NDVI to localize infected areas and estimate the percentage of leaf damage. Environmental parameters such as temperature, humidity, and soil moisture can further enhance prediction models, as these conditions directly influence disease development and spread.

This research builds on these advancements by developing a hybrid system that combines CNN-based image classification, NDVI-based vegetation health assessment, and environmental parameter analysis for black pepper disease detection and spread quantification. By doing so, it addresses key gaps in current literature, which often lack robust, integrated approaches for black pepper disease management that go beyond simple classification to include early physiological detection and actionable severity metrics.

1.2 Research Gap

The field of plant disease detection has undergone significant progress in recent years with the application of machine learning and deep learning models. CNN-based methods, in particular, have demonstrated high levels of accuracy for crops such as rice, maize, cucumber, and tomato [1], [2], [4], [7], [14]. These studies have shown that image-based recognition can be an effective solution for automated diagnosis. However, research on black pepper (*Piper nigrum* L.), a high-value spice crop cultivated extensively in tropical regions, remains considerably underdeveloped [8].

Existing black pepper disease studies rely predominantly on RGB image-based CNN classification. Although effective in identifying visible disease symptoms, these methods face three key limitations:

1. Absence of physiological stress detection

RGB image classifiers depend on visible lesions, spots, or discoloration, meaning they can only recognize disease after significant progression. Early physiological changes such as reduced chlorophyll levels or weakened photosynthetic activity cannot be captured through visible-spectrum imaging alone [5], [6]. In contrast, spectral indices such as the Normalized Difference Vegetation Index (NDVI) have been proven effective for detecting stress prior to visible symptom expression [5], [9]. Nevertheless, in black pepper disease research, NDVI has been rarely applied, and when used, it typically serves as a standalone metric without integration into classification frameworks. This creates a diagnostic blind spot, as early-stage detection remains largely unaddressed.

2. Minimal emphasis on severity quantification

Most CNN-based studies classify leaves into broad categories (e.g., healthy, mild, severe), which provides limited information for field-level management [3]. Farmers and agronomists often require precise quantification of infection, such as the proportion of leaf area affected, in order to prioritize treatments and manage

resources effectively. Image segmentation methods like U-Net [10] and Mask R-CNN [11] offer potential for pixel-level analysis, yet their application to black pepper disease quantification remains scarce. Without quantitative severity estimation, existing models lack practical utility for large-scale disease monitoring.

3. Limited adaptability to real-world agricultural conditions

Models developed in controlled laboratory environments often fail in field conditions, where factors such as variable sunlight, overlapping leaves, soil background interference, and irregular camera angles affect detection reliability [4], [15]. Additionally, environmental factors such as temperature, humidity, and soil moisture are known to influence disease development and progression [12]. Despite this, few black pepper-specific studies incorporate environmental variables into predictive models, thereby limiting context-aware diagnosis and forecasting capabilities [13].

4. Underutilization of multimodal integration

Current approaches often treat disease detection as either an image classification problem or a vegetation index-based stress assessment problem. This separation reduces diagnostic accuracy. NDVI provides early physiological insights but cannot distinguish between disease-induced stress and abiotic stressors such as nutrient deficiencies or drought [6]. Conversely, CNN models can classify visible disease types but cannot detect pre-symptomatic stress. This disconnection restricts the effectiveness of both methods when used independently. A combined framework that fuses RGB images, NDVI data, and environmental parameters could bridge this gap, enhancing both early detection and diagnostic specificity [9], [12], [13].

5. Lack of dedicated datasets for black pepper disease

Widely used datasets such as PlantVillage are invaluable for CNN model training [7], [14]. However, they lack black pepper leaf images, NDVI readings, and associated environmental data. Existing literature on black pepper pathology in Sri

Lanka provides valuable information on disease identification and management practices [8], but no standardized dataset currently exists that integrates multimodal inputs necessary for hybrid models. This scarcity of data represents a major barrier to developing and validating robust black pepper disease detection systems.

Summary of the Gap

From the reviewed literature, it is evident that while CNN-based disease classification and NDVI-based stress monitoring have shown promise in agricultural research, their application to black pepper is fragmented and insufficient. The primary gaps include:

- Absence of **early-stage physiological stress detection** in black pepper research.
- Lack of **precise severity quantification** using segmentation methods.
- Poor **adaptability to field conditions** with minimal integration of environmental parameters.
- Insufficient efforts toward **multimodal integration** of CNN, NDVI, and IoT-based environmental monitoring.
- **Scarcity of dedicated datasets** for black pepper disease under real-world tropical agricultural conditions.

Proposed Solution

This research addresses these gaps by introducing a **hybrid detection framework** that integrates:

1. CNN-based RGB classification for symptom detection [1], [2]
2. NDVI scanning for early physiological stress identification [5], [6]
3. Image segmentation (U-Net, Mask R-CNN) for quantitative disease spread analysis [10], [11]

4. Environmental parameter integration for context-aware risk forecasting [12], [13].

By combining these complementary approaches, the proposed system aims to achieve early disease detection, accurate classification of specific pathogens, precise quantification of infection severity, and predictive analysis under real-world tropical field conditions. Such a system would provide farmers with a reliable decision-support tool to improve disease management strategies, reduce crop losses, and safeguard black pepper yield and quality.

Feature	Existing Research on Crops (Rice, Maize, Tomato, etc.)	Existing Research on Black Pepper	Proposed System
Early-stage stress detection (NDVI / spectral indices)	✓ Widely applied for rice, maize, etc. [5], [9]	✗ Rarely applied, mostly absent in pepper studies	✓ Integrated NDVI scanning for early stress
Precise severity quantification (leaf area %)	✓ Applied in some crops using U-Net / Mask R-CNN [10], [11]	✗ Pepper research mostly limited to categorical labels (healthy, mild, severe)	✓ Segmentation-based % severity estimation
Adaptability to real field conditions (sunlight, soil, overlaps)	✗ Many CNN models fail in uncontrolled environments [4], [15]	✗ Pepper-specific models rarely incorporate environmental factors	✓ Incorporates temperature, humidity, soil moisture for context
Multimodal integration (CNN + NDVI + Environment)	✗ Most studies treat CNN and NDVI separately	✗ Black pepper research does not integrate multimodal data	✓ Unified CNN + NDVI + IoT framework
Availability of datasets (black pepper specific)	✓ Global datasets like PlantVillage [7], [14] (but exclude pepper)	✗ Very limited datasets; no multimodal black pepper dataset	✓ New dataset including RGB, NDVI, and environment parameters

1.3 Research Problem

Black pepper cultivation plays a vital role in the agricultural economy of many tropical countries, including Sri Lanka. However, the crop is highly vulnerable to a variety of leaf diseases such as *Phytophthora foot rot*, anthracnose, and yellow mottle virus. These diseases often spread rapidly, especially under favorable environmental conditions, and can cause significant yield losses if not detected and treated early.

The main challenge lies in **accurately detecting these diseases in their early stages** and **quantifying the extent of infection** to guide timely and effective intervention. Conventional detection methods rely on visual inspection by agricultural experts, which, while practical, is:

- **Time-consuming** and requires trained personnel.
- **Subjective**, as early symptoms can be easily confused with nutrient deficiencies, water stress, or natural leaf aging.
- **Reactive rather than proactive**, as by the time symptoms are clearly visible, the disease may already be widespread.

Existing automated disease detection research has primarily focused on **RGB image-based Convolutional Neural Networks (CNNs)** to identify visible symptoms. While effective for classification, such approaches cannot detect pre-symptomatic stress and often lack the capability to estimate the **percentage of infected area** accurately. This limitation reduces their usefulness for severity assessment and disease management planning.

On the other hand, the **Normalized Difference Vegetation Index (NDVI)** has shown strong potential for detecting plant stress at a physiological level before visible symptoms appear, by analyzing changes in light reflectance patterns. However, NDVI alone cannot differentiate between specific diseases and other stress factors, limiting its diagnostic value.

The research problem, therefore, is:

How can we design an integrated detection system that combines CNN-based image analysis, NDVI-based physiological assessment, and environmental parameter

monitoring to enable early, accurate detection and quantification of black pepper leaf diseases in real-world field conditions?

Such a system must be able to:

1. **Detect early physiological stress** using NDVI scanning, even before symptoms are visible.
2. **Identify the specific disease type** through CNN-based RGB image classification.
3. **Quantify disease spread** using image segmentation to measure infected areas.
4. **Predict potential outbreaks** by correlating detection results with environmental conditions such as humidity, temperature, and soil moisture.

Addressing this problem would allow for **faster, more accurate, and more objective decision-making** in black pepper disease management, enabling farmers to implement targeted treatments, reduce unnecessary pesticide usage, and ultimately safeguard crop yield and quality.

2. OBJECTIVES

The main objective of this research is to develop a robust and integrated system for the detection and spread analysis of black pepper leaf diseases. The system will combine CNN-based image analysis, NDVI-based stress detection, image segmentation, and environmental parameter monitoring to enable early and accurate disease management in real-world agricultural settings.

2.1 Main Objectives

The primary objective of this research is to design and implement an automated detection framework that:

- **Identifies black pepper leaf diseases accurately** using CNN-based classification models.
- **Detects early-stage physiological stress** using NDVI values before visible symptoms appear.
- **Quantifies infection severity** by calculating the percentage of infected leaf area and estimating whole-plant disease spread.
- **Predicts disease risk** through the integration of environmental parameters such as temperature, humidity, and soil moisture.

By achieving these goals, the proposed system aims to provide farmers and agricultural officers with a reliable decision-support tool that reduces crop loss, minimizes pesticide misuse, and supports sustainable black pepper production.

2.2 Specific Objectives

To achieve the main objective, the study will pursue the following specific objectives:

1. **Develop a comprehensive dataset** of black pepper leaves, including healthy and diseased samples, captured under diverse lighting and environmental conditions. This dataset will include both RGB images and NDVI measurements.
2. **Design and train CNN-based models** capable of classifying black pepper diseases from RGB images with high accuracy, even under field conditions with variable lighting and background clutter.

3. **Integrate NDVI-based analysis** to detect early physiological stress in black pepper plants and correlate NDVI values with the onset of visible disease symptoms.
4. **Apply image segmentation techniques** (e.g., U-Net, Mask R-CNN) to localize diseased regions on leaves and calculate the percentage of infection, enabling severity estimation at both leaf and plant levels.
5. **Incorporate environmental parameter monitoring** (temperature, humidity, and soil moisture) into the system to model disease spread patterns and predict outbreak risks under varying field conditions.
6. **Benchmark the system's performance** against expert agricultural assessments to evaluate accuracy, robustness, and practicality in real-world farming environments.
7. **Provide a user-friendly output system** that generates reports and visualizations for farmers and agricultural officers, including disease type, severity percentage, NDVI-based stress indicators, and risk predictions.

3. METHODOLOGY

3.1 System Overview

The proposed system is designed to provide early detection, classification, and spread analysis of black pepper leaf diseases using a hybrid approach that combines image-based deep learning, NDVI scanning, and environmental parameter monitoring. The system workflow involves four main steps:

1. **Data Collection** – Capturing RGB images of black pepper leaves, NDVI sensor data, and environmental parameters such as temperature, humidity, and soil moisture from both field and controlled environments.
2. **Preprocessing** – Enhancing image quality, segmenting leaf regions, and removing noise from RGB and NDVI data. Environmental data will also be cleaned and standardized for analysis.

3. **Disease Detection & Analysis** – Using CNN-based classification to identify disease types, NDVI values to detect early physiological stress, and image segmentation models (e.g., U-Net, Mask R-CNN) to estimate infection severity.
4. **Prediction & Reporting** – Integrating disease detection results with environmental data to predict disease risk, generate severity reports, and provide farmers with actionable insights through a decision-support dashboard.

This hybrid framework ensures that the system can detect early stress, classify disease types accurately, and quantify infection spread, thereby enabling timely intervention and effective disease management.

3.2 System Architecture

The architecture of the proposed system consists of five core components:

1. **Image Acquisition Module** – Collects RGB images of black pepper leaves using cameras or smartphones under field conditions.
2. **NDVI Sensing Module** – Captures NDVI values from handheld or drone-mounted sensors to measure vegetation health.
3. **Environmental Monitoring Module** – Records temperature, humidity, and soil moisture levels using IoT-based sensors.
4. **Analysis Engine** –
 - CNN-based classifier for disease type recognition.
 - Image segmentation model for severity estimation.
 - NDVI analyzer for early stress detection.
 - Data fusion algorithm that integrates image, NDVI, and environmental data.
5. **Decision-Support Interface** – Provides users with disease identification, severity percentage, NDVI health index, outbreak risk predictions, and recommendations through an interactive dashboard.

3.2.1 Individual Components

- **CNN-based Classification:** A deep learning model trained on the black pepper leaf dataset to classify diseases such as foot rot, anthracnose, and yellow mottle virus. Data augmentation will be applied to improve performance under variable field conditions.
- **NDVI Analysis:** NDVI values will be computed from multispectral reflectance data. These values will be used to detect early stress by correlating NDVI with chlorophyll content and disease progression.
- **Image Segmentation:** U-Net or Mask R-CNN models will be applied to localize diseased regions on leaves and calculate the percentage of infected area, providing a quantitative measure of severity.
- **Environmental Data Integration:** Temperature, humidity, and soil moisture will be analyzed alongside disease data to model conditions favorable for disease outbreaks. Predictive models (e.g., regression or time-series forecasting) will be used for risk assessment.
- **Decision-Support System:** Outputs a combined report including disease type, severity percentage, NDVI stress indicators, and environmental risk predictions. The report will be visualized using graphs, heatmaps, and NDVI health maps for easy interpretation.

3.3 Software Solution

The software solution will be developed using a combination of AI frameworks and IoT technologies:

- **Deep Learning & Image Processing:** TensorFlow, PyTorch, OpenCV.
- **NDVI Processing:** Python libraries for spectral index calculation, integration with multispectral cameras or sensors.
- **Segmentation Models:** U-Net and Mask R-CNN for infection spread quantification.

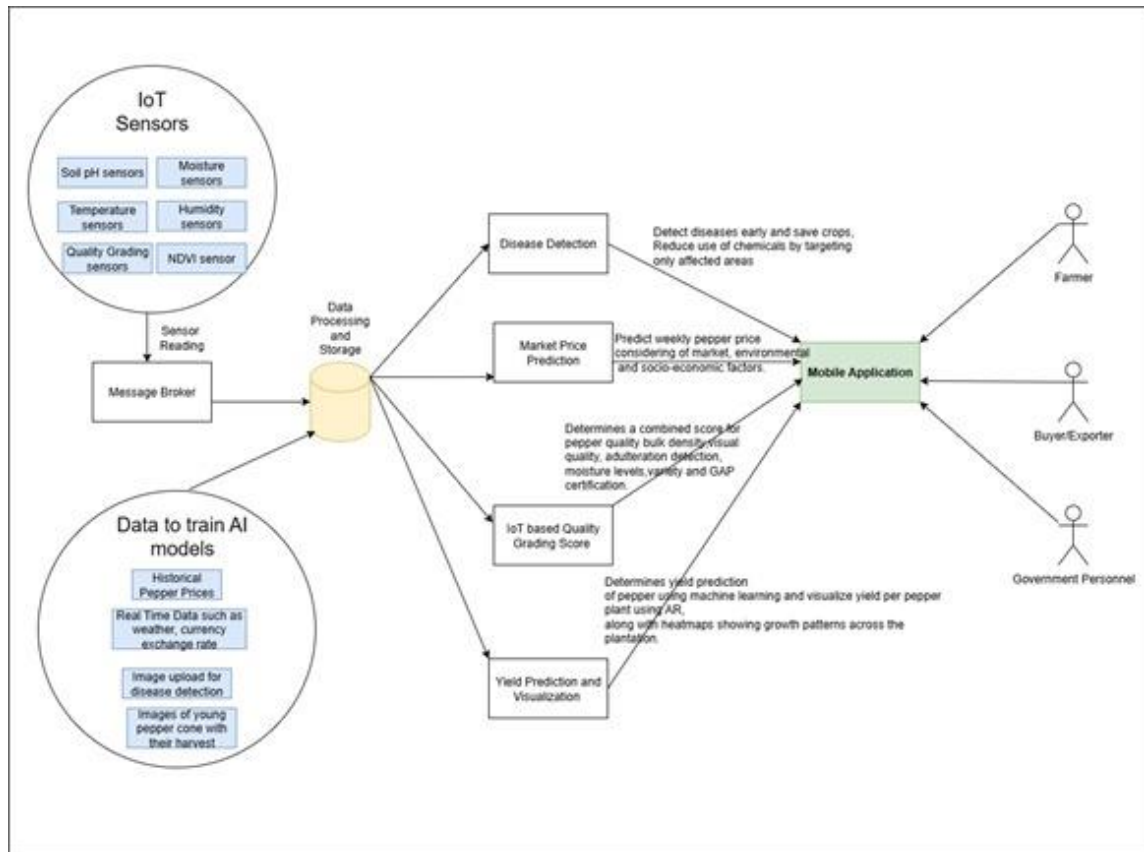
- **Environmental Data Handling:** IoT data collection via Arduino/Raspberry Pi, storage in MongoDB, preprocessing in Python (Pandas, NumPy).
- **Integration & Dashboard:** Web-based interface developed using React.js, Node/Express backend, and charting tools for visualization.

Agile methodology will be followed for iterative development, allowing continuous testing and refinement of the system.

3.4 Future Scope

While this research focuses on CNN, NDVI, and environmental parameters for disease detection and spread analysis in black pepper, future extensions may include:

- **Integration with drone-based multispectral imaging** for large-scale field monitoring.
- **Inclusion of other vegetation indices** (e.g., GNDVI, SAVI) to improve stress detection accuracy.
- **Deployment as a mobile application** to make disease detection accessible to farmers with smartphones.
- **Integration with early warning systems** for predicting regional disease outbreaks.
- **Expansion to other spice crops** such as cinnamon and cardamom, using transfer learning on the developed models.



4. PROJECT REQUIREMENTS

4.1 Functional Requirements

The system must provide the following core functionalities:

1. Data Collection

- Capture RGB images of healthy and diseased black pepper leaves using field cameras or smartphones.

- Collect NDVI values using handheld or drone-mounted multispectral sensors.
- Monitor environmental parameters (temperature, humidity, soil moisture) using IoT sensors.

2. Data Preprocessing

- Enhance images (noise removal, normalization, segmentation).
- Filter and standardize NDVI and environmental sensor readings.
- Remove irrelevant data (e.g., background clutter, shadow distortions).

3. Disease Detection & Analysis

- CNN-based classification of leaf diseases.
- NDVI-based detection of early physiological stress.
- Image segmentation to estimate the percentage of infection spread.

4. Integration of Multi-Modal Data

- Combine CNN results, NDVI values, and environmental conditions to provide comprehensive disease analysis.
- Correlate environmental trends with disease progression for risk prediction.

5. Reporting & Visualization

- Generate disease diagnosis reports including:
 - Disease type
 - Severity percentage (infected area)
 - NDVI-based vegetation health index
 - Environmental conditions and outbreak risk level
- Display results on a web dashboard with graphs, heatmaps, and NDVI maps.

4.2 Non-Functional Requirements

- **Performance:** The system must process images and sensor data in near real-time (under 5 seconds per leaf image).
- **Accuracy:** CNN classification and segmentation accuracy should exceed 85% under field conditions.
- **Scalability:** Capable of handling large datasets from multiple farms or drone-based image collections.
- **Security:** All sensor and image data must be securely stored and transmitted with encryption.
- **Reliability:** Must provide consistent disease detection under variable lighting and environmental conditions.
- **Usability:** Farmers and agricultural officers should be able to operate the system with minimal training.
- **Maintainability:** System should allow easy model retraining with new datasets for improved accuracy.

4.3 System Requirements

Hardware Requirements:

- RGB Camera or smartphone with HD resolution.
- Multispectral sensor or NDVI handheld scanner / drone-based sensor.
- IoT sensors for temperature, humidity, and soil moisture.
- Local server or cloud storage for dataset management.
- GPU-enabled machine for CNN training (e.g., NVIDIA RTX series).

Software Requirements:

- **AI & Image Processing:** TensorFlow, PyTorch, OpenCV.
- **NDVI Processing:** Python libraries for spectral index analysis.
- **Segmentation:** U-Net, Mask R-CNN frameworks.

- **Data Handling:** MongoDB / MySQL for storage, Pandas & NumPy for preprocessing.
- **Dashboard:** React.js (frontend), Node/Express (backend).

Networking:

- Stable internet connectivity for real-time cloud processing and data transmission.
- IoT device integration with Wi-Fi or LoRaWAN for environmental sensor data.

4.4 User Requirements

- **Ease of Use:** Farmers should be able to upload images and receive results without technical expertise.
- **Real-time Interaction:** System should provide near-instant results for leaf scans and NDVI readings.
- **Customizable Analysis:** Ability to configure detection for different diseases and severity levels.
- **Offline Support:** System should allow offline scanning with later synchronization to cloud when internet is available.
- **Data Privacy:** Ensure user data, especially field location and farm conditions, is securely stored and not shared without consent.
- **Actionable Insights:** Reports must include recommendations for next steps (e.g., apply fungicide, isolate infected plants, monitor environmental conditions).
- **Support & Training:** Provide user manuals or short tutorials for farmers and agricultural officers.

5. BUDGET AND BUDGET JUSTIFICATION

The proposed system requires both hardware and software resources for data collection, model development, system integration, and deployment. The estimated costs are outlined below:

Requirement	Estimated Cost (LKR)	Justification
RGB Camera / Smartphone (HD)	30,000	For capturing high-quality images of black pepper leaves under field conditions.
NDVI Sensor (Handheld / Drone-based)	6,000	For scanning vegetation health and detecting early physiological stress.
IoT Sensors (Temperature, Humidity, Moisture)	2,500	To continuously record environmental parameters influencing disease spread.
GPU-enabled Machine (Local / Cloud)	25,000	Required for training CNN and segmentation models efficiently.
Server / Cloud Hosting	20,000	For storing large datasets and hosting the backend system.
Deployment (Frontend + Backend + DB)	15,000	Setting up the system for testing and user accessibility via dashboard.
Miscellaneous (Internet, Power, Maintenance)	5,000	Covers internet usage, electricity, and maintenance during development.
Total Estimated Cost: 103,500 LKR		

Budget Justification

- **Data Collection:** Cameras and NDVI sensors are essential for building a reliable dataset of diseased and healthy black pepper leaves, along with vegetation health indices. IoT sensors will provide environmental parameters to improve disease prediction models.
- **Model Development:** GPU-enabled hardware (local or cloud) is required to handle deep learning model training (CNNs, U-Net, Mask R-CNN), which is computationally expensive.
- **System Deployment:** Cloud hosting ensures scalability and accessibility for farmers and agricultural officers. Deployment costs include setting up the web dashboard and backend services.
- **Operational Costs:** Internet connectivity, electricity, and maintenance are necessary to support data transfer, sensor operation, and long-term usage of the system.

The allocated budget ensures that the proposed system is feasible, sustainable, and capable of delivering accurate, real-time disease detection and analysis for black pepper crops.

Grant chart



7. DESCRIPTION OF PERSONNEL AND FACILITIES

Personnel

- **Researcher (Undergraduate Student)**
 - Responsible for overall project design, dataset collection, model development, integration of CNN, NDVI, and environmental data, and preparation of the final dissertation.
 - Tasks include literature review, data preprocessing, training deep learning models, building the decision-support system, and evaluating system performance.
- **Supervisor (Academic Staff – SLIIT)**
 - Provides academic guidance, technical advice, and support throughout the research project.
 - Ensures the research meets scientific and ethical standards, and reviews progress at each milestone.
- **Co-Supervisor / Domain Expert**
 - Offers subject matter expertise in agriculture and plant pathology, particularly black pepper diseases.
 - Assists in validating disease datasets and ensuring the accuracy of classification and severity analysis.
- **External Collaborators**

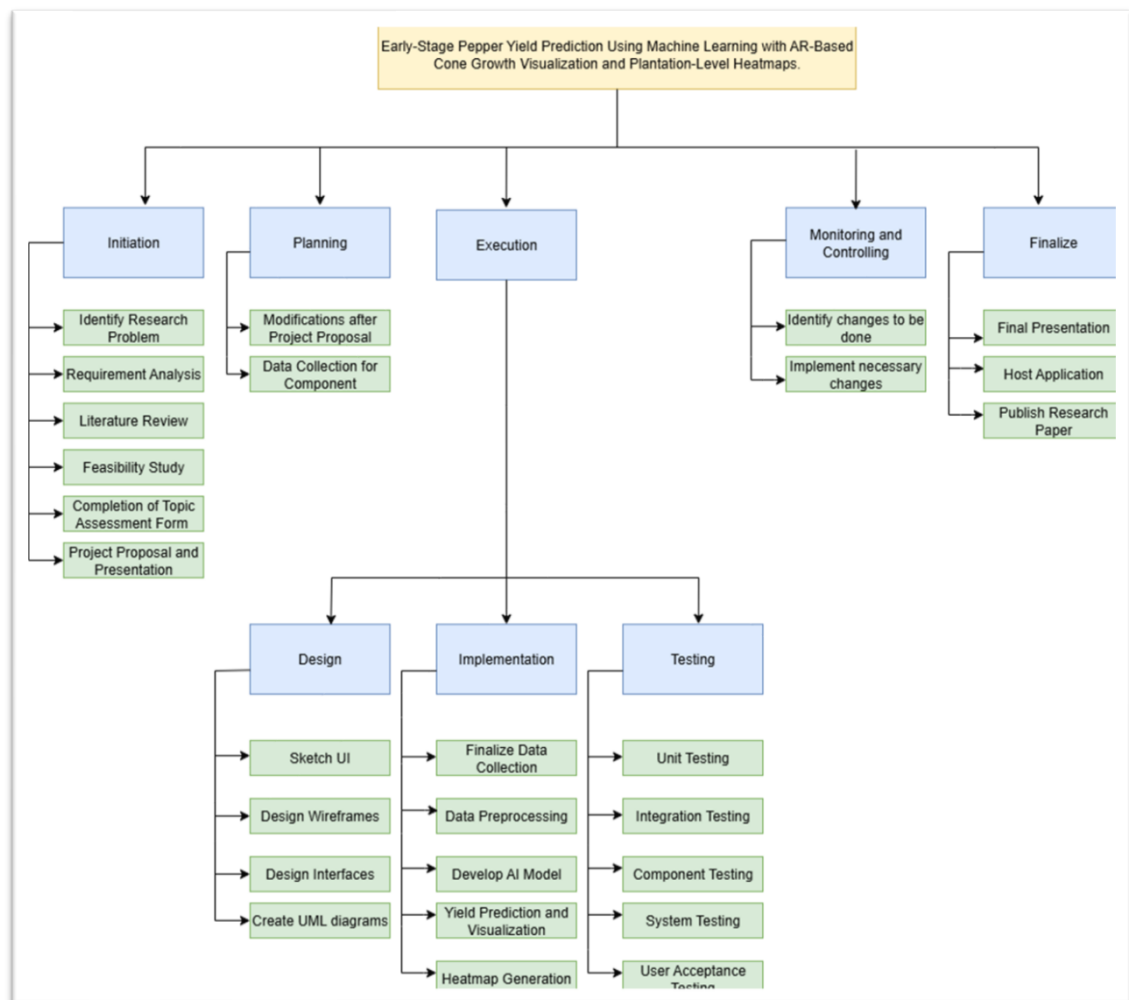
- Agricultural officers or local farmers provide access to black pepper fields for data collection.
- May assist in ground-truth validation by confirming disease type and severity in real-world conditions.

Facilities

- **Sri Lanka Institute of Information Technology (SLIIT) Facilities**
 - Computer laboratories equipped with GPU-enabled machines for CNN training and model development.
 - Access to academic software licenses and research resources.
- **Agricultural Field Sites**
 - Black pepper plantations in Sri Lanka will be used for real-world data collection.
 - Field sites allow testing of NDVI sensors, IoT devices, and mobile deployment of the system under practical farming conditions.
- **Research Tools and Equipment**
 - RGB camera / smartphone for capturing leaf images.
 - NDVI handheld sensor or drone-mounted multispectral sensor for vegetation health scanning.
 - IoT sensor kits (temperature, humidity, soil moisture) for monitoring environmental factors.
- **Cloud Computing and Storage**
 - Cloud servers for storing image datasets, NDVI readings, and environmental data.

- GPU-enabled cloud platforms (e.g., Google Colab, AWS, or Azure) for deep learning training and testing.

WORK BREAKDOWN STRUCTURE



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